

CE 310 — Week 5 Pre-Lab Quiz

Topic: Normal Distribution — Concepts, Excel Functions, and Engineering Applications
Points: 10 questions $\times$ 1 pt = 10 pts Due: Before class begins — submit in D2L.

Questions

Q1. The Normal distribution is fully defined by ____ parameter(s).

- (A) One: the mean μ
 - (B) Two: μ and σ
 - (C) Three: mean, standard deviation, and sample size
 - (D) Four: mean, standard deviation, skewness, and kurtosis
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Q2. In Excel, =NORM.DIST(x , μ , σ , TRUE) computes:

- (A) $P(X \leq x)$ — the cumulative probability up to x
 - (B) $P(X = x)$ — the probability density at a single point
 - (C) $P(X > x)$ — the exceedance probability above x
 - (D) The z-score for x
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Q3. =NORM.INV(0.10, μ , σ) returns:

- (A) The z-score at the 10th percentile (always -1.282)
 - (B) The probability that X is less than 0.10
 - (C) The value x below which 10% of observations fall
 - (D) The value x above which 10% of observations fall
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Q4. Approximately what percentage of observations fall within 2 standard deviations of the mean for a Normal distribution?

- (A) 68%
 - (B) 95%
 - (C) 99.7%
 - (D) 50%
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Q5. Concrete compressive strength measurements from your dataset have $\mu = 4,496$ psi and $\sigma = 442$ psi. Using only the 68–95–99.7 rule (no Excel), which answer best estimates $P(f_c < 4,000 \text{ psi})$?

- (A) $z \approx -1.12$; estimated probability is roughly 10–14%

- (B) $z \approx -1.12$; estimated probability is roughly 50%
 - (C) $z \approx +1.12$; estimated probability is roughly 87%
 - (D) $z \approx -2.0$; estimated probability is roughly 2.5%
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Q6. ACI 318 requires that no more than ____% of concrete test cylinders fall below the specified compressive strength f'_c .

- (A) 5%
 - (B) 10%
 - (C) 15%
 - (D) 20%
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Q7. The 85th percentile speed is the standard engineering metric for evaluating posted speed limits. Traffic spot speeds on a study corridor have $\mu = 54.4$ mph and $\sigma = 7.8$ mph. Which Excel formula gives the 85th percentile design speed?

- (A) =NORM.INV(0.85, 54.4, 7.8)
 - (B) =NORM.DIST(0.85, 54.4, 7.8, TRUE)
 - (C) =NORM.INV(0.15, 54.4, 7.8)
 - (D) =NORM.INV(0.50, 54.4, 7.8)
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Q8. A lognormal distribution is most appropriate when data:

- (A) Is symmetric and ranges from $-\infty$ to $+\infty$
 - (B) Is right-skewed, always positive, and whose natural log is approximately Normal
 - (C) Is left-skewed and bounded above
 - (D) Counts discrete events over a fixed interval
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Q9. The Normal distribution appears frequently in engineering measurements because:

- (A) Engineers prefer simple two-parameter models
 - (B) Measured quantities are often the sum of many small, independent influences — a consequence of the Central Limit Theorem
 - (C) The Normal distribution fits all data equally well
 - (D) Engineering standards require using the Normal distribution
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Q10. In your dataset, Supplier Alpha has $\sigma = 344$ psi and Supplier Beta has $\sigma = 484$ psi. Both suppliers target the same specified $f'_c = 4,000$ psi. Which statement is correct?

- (A) Alpha has a higher failure rate because its σ is smaller
- (B) Beta is more consistent because its σ is larger

- (C) Alpha is more likely to meet ACI 318's 10% failure limit because its σ is smaller and tighter around a higher mean
 - (D) The failure rate depends only on the mean, not σ
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